Wireshark Workshop

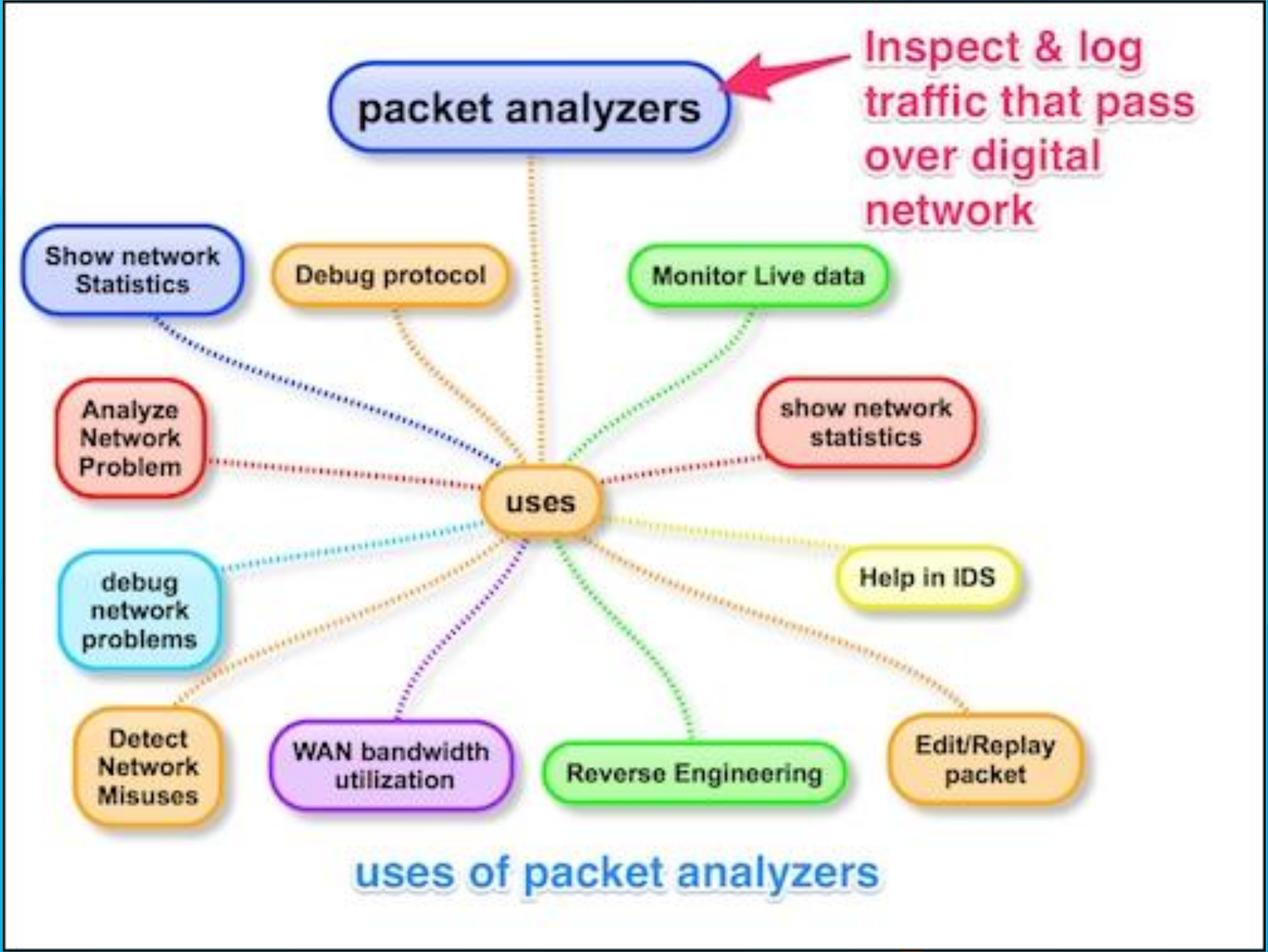
ACM SIG Cyber

Packet Analyzers

- Also called “Packet Sniffer” has the ability to grab the raw packet from the wire, wireless, Bluetooth, VLAN, PPP, and other network types without getting processed by the application. We’ll cover:
    - Uses for packet sniffers
    - Introducing Wireshark
    - Other packet sniffer tools
    - Mobile Packet Capturing

Uses for Packet Analyzers

Network Security: Analyze raw network traffic to detect scans and attacks, for sniffing, network troubleshooting, and many more!

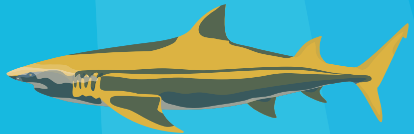

packet analyzers
Inspect & log traffic that pass over digital network
Show network Statistics
Debug protocol
Monitor Live data
Analyze Network Problem
uses
show network statistics
debug network problems
Help in IDS
Detect Network Misuses
WAN bandwidth utilization
Reverse Engineering
Edit/Replay packet
uses of packet analyzers

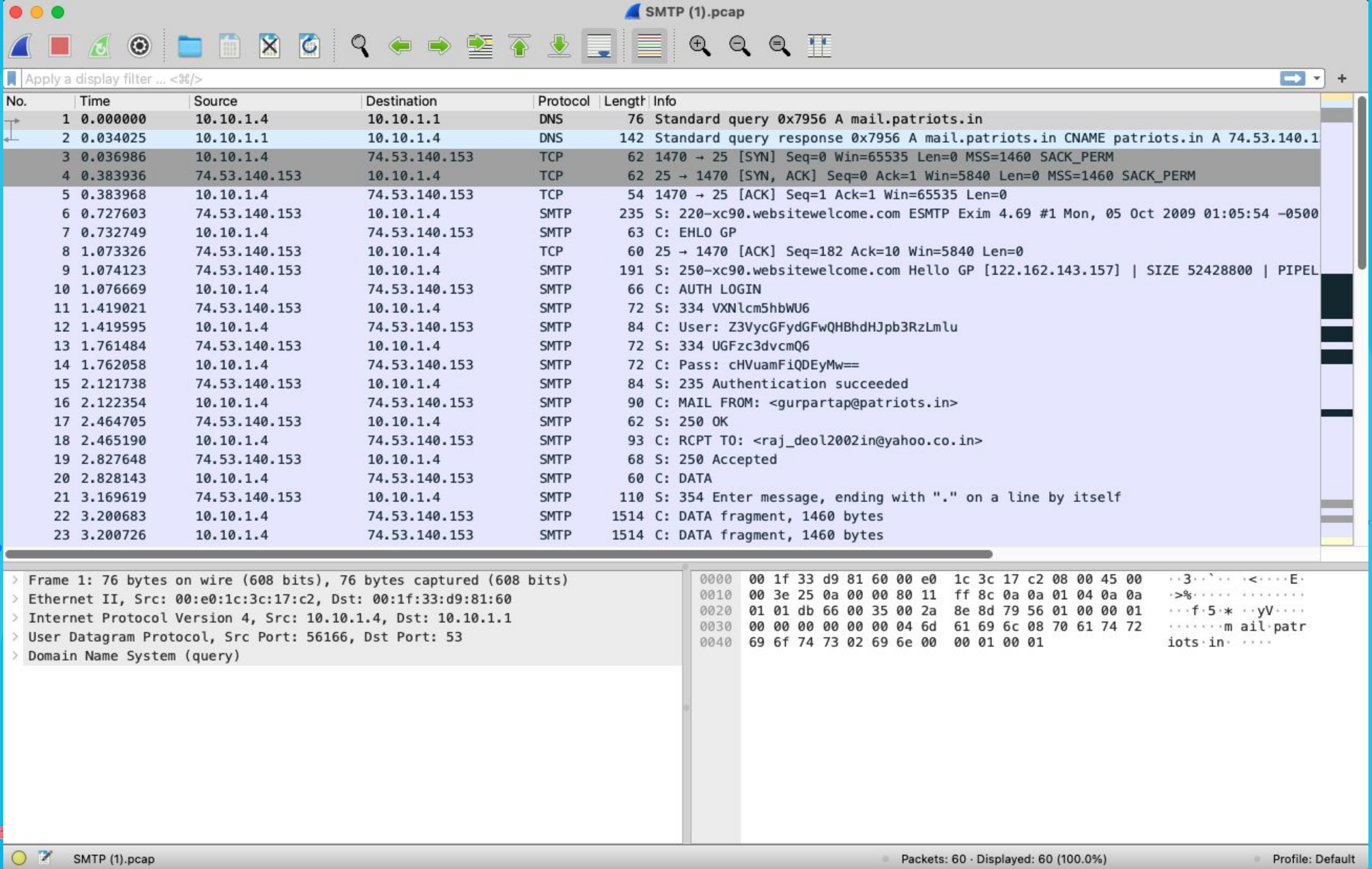
Example Uses for Packet Analyzers

- Network Administrators troubleshooting problems on a network
- Security Architects can perform a security audit on a packet
- Developers can use it to debug protocol implementations
- White-hat hackers can find vulnerabilities in the application and fix them before black-hat hackers can find them

Introducing Wireshark

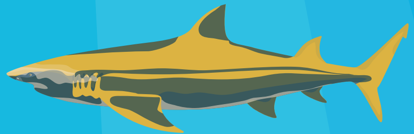
- One of the best open-source network packet analyzers
  - Powerful packet analyzer tool with an easy-to-use GUI and command-line
- Wireshark uses pcap files to capture packets, meaning it can capture packets in offline mode (to view the captured packets) and online mode (live traffic).

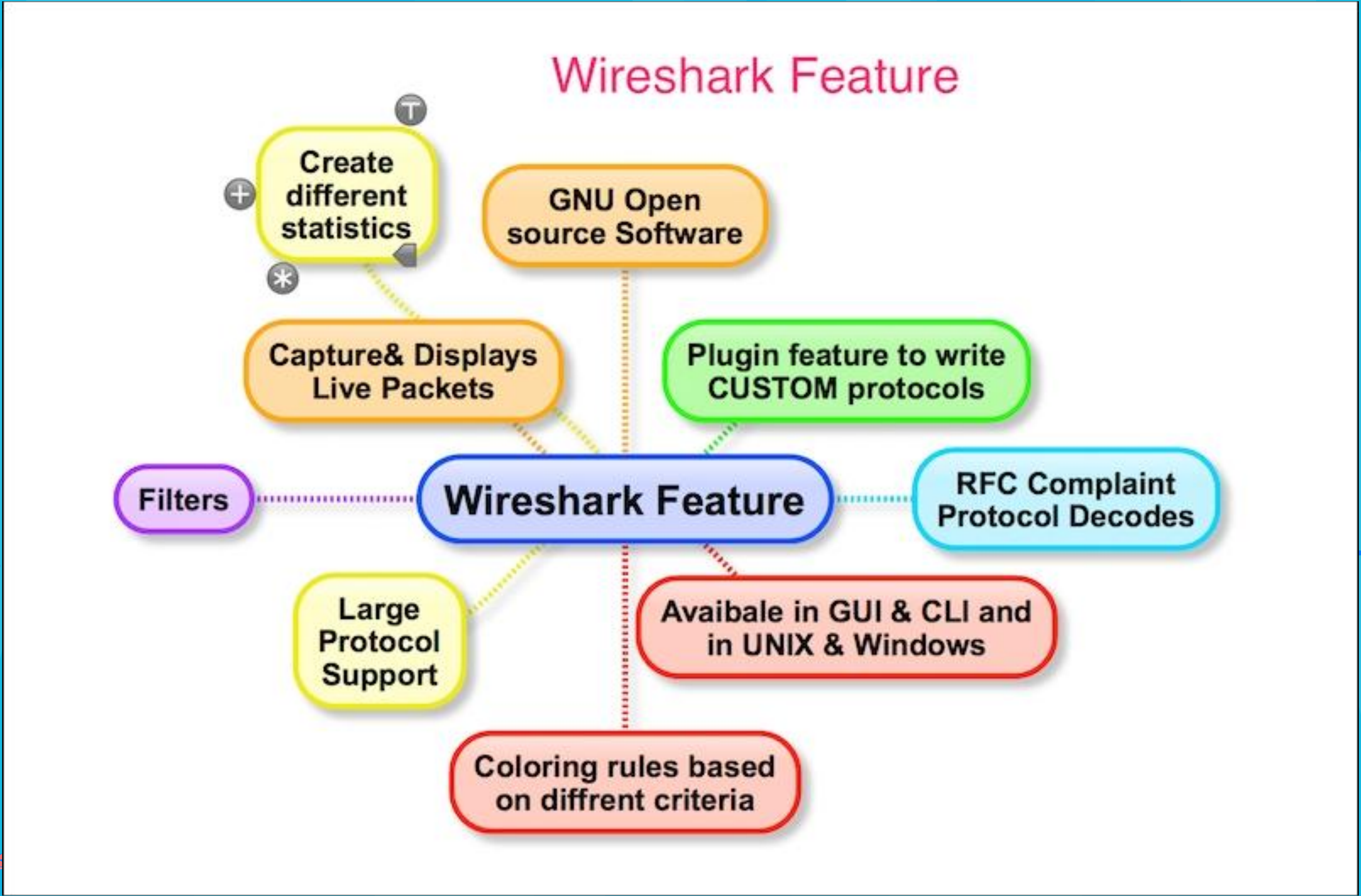



SMTP (1).pcap
Apply a display filter ... <⌘/>
No. Time Source Destination Protocol Length Info
1 0.000000 10.10.1.4 10.10.1.1 DNS 76 Standard query 0x7956 A mail.patriots.in
2 0.034025 10.10.1.1 10.10.1.4 DNS 142 Standard query response 0x7956 A mail.patriots.in CNAME patriots.in A 74.53.140.1
3 0.036986 10.10.1.4 74.53.140.153 TCP 62 1470 → 25 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM
4 0.383936 74.53.140.153 10.10.1.4 TCP 62 25 → 1470 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460 SACK_PERM
5 0.383968 10.10.1.4 74.53.140.153 TCP 54 1470 → 25 [ACK] Seq=1 Ack=1 Win=65535 Len=0
6 0.727603 74.53.140.153 10.10.1.4 SMTP 235 S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
7 0.732749 10.10.1.4 74.53.140.153 SMTP 63 C: EHLO GP
8 1.073326 74.53.140.153 10.10.1.4 TCP 60 25 → 1470 [ACK] Seq=182 Ack=10 Win=5840 Len=0
9 1.074123 74.53.140.153 10.10.1.4 SMTP 191 S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] | SIZE 52428800 | PIPEL
10 1.076669 10.10.1.4 74.53.140.153 SMTP 66 C: AUTH LOGIN
11 1.419021 74.53.140.153 10.10.1.4 SMTP 72 S: 334 VXNlcm5hbWU6
12 1.419595 10.10.1.4 74.53.140.153 SMTP 84 C: User: Z3VycGFydGFwQHBhdHJpb3RzLmlu
13 1.761484 74.53.140.153 10.10.1.4 SMTP 72 S: 334 UGFzc3dvcmQ6
14 1.762058 10.10.1.4 74.53.140.153 SMTP 72 C: Pass: cHVuamFiQDEyMw==
15 2.121738 74.53.140.153 10.10.1.4 SMTP 84 S: 235 Authentication succeeded
16 2.122354 10.10.1.4 74.53.140.153 SMTP 90 C: MAIL FROM: <gurpartap@patriots.in>
17 2.464705 74.53.140.153 10.10.1.4 SMTP 62 S: 250 OK
18 2.465190 10.10.1.4 74.53.140.153 SMTP 93 C: RCPT TO: <raj_deol2002in@yahoo.co.in>
19 2.827648 74.53.140.153 10.10.1.4 SMTP 68 S: 250 Accepted
20 2.828143 10.10.1.4 74.53.140.153 SMTP 60 C: DATA
21 3.169619 74.53.140.153 10.10.1.4 SMTP 110 S: 354 Enter message, ending with "." on a line by itself
22 3.200683 10.10.1.4 74.53.140.153 SMTP 1514 C: DATA fragment, 1460 bytes
23 3.200726 10.10.1.4 74.53.140.153 SMTP 1514 C: DATA fragment, 1460 bytes
Frame 1: 76 bytes on wire (608 bits), 76 bytes captured (608 bits)
Ethernet II, Src: 00:e0:1c:3c:17:c2, Dst: 00:1f:33:d9:81:60
Internet Protocol Version 4, Src: 10.10.1.4, Dst: 10.10.1.1
User Datagram Protocol, Src Port: 56166, Dst Port: 53
Domain Name System (query)
0000 00 1f 33 d9 81 60 00 e0 1c 3c 17 c2 08 00 45 00
0010 00 3e 25 0a 00 00 80 11 ff 8c 0a 0a 01 04 0a 0a
0020 01 01 db 66 00 35 00 2a 8e 8d 79 56 01 00 00 01
0030 00 00 00 00 00 00 04 6d 61 69 6c 08 70 61 74 72
0040 69 6f 74 73 02 69 6e 00 00 01 00 01
SMTP (1).pcap
Packets: 60 · Displayed: 60 (100.0%)
Profile: Default

Wireshark Features

- Available in both UNIX and Windows
- Ability to capture live packets from various types of interface
- Filter packets with many criteria
- Ability to decode larger sets of protocols
- Can save and merge captured packets
- Can create various statistics




Wireshark Feature
Create different statistics
GNU Open source Software
Capture& Displays Live Packets
Plugin feature to write CUSTOM protocols
Filters
Wireshark Feature
RFC Complaint Protocol Decodes
Large Protocol Support
Avaibale in GUI & CLI and in UNIX & Windows
Coloring rules based on diffrent criteria

Wireshark's dumpcap/tshark

- Wireshark provides command-line tools "dumpcap" and "tshark" What the heck are those????
    - Dumpcap: Dump network traffic
        - Capture packet data from a live network and write the packets to a file in pcapng format. When the -P option is specified, the output file will be written in the pcap format.

- Wireshark & tshark -> dumpcap

More in depth

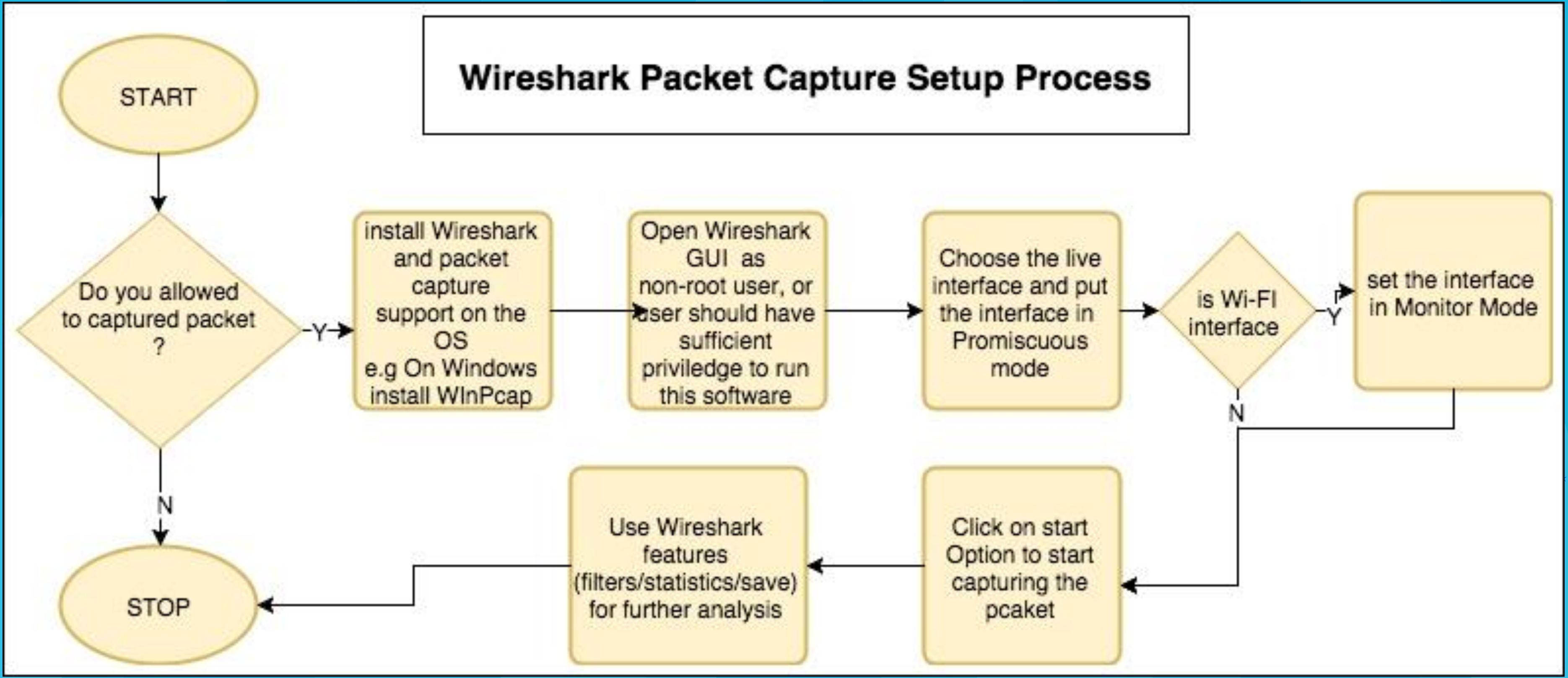
- Wireshark is a graphical application
- tshark is that application without the GUI
- dumpcap, per Wireshark's documentation, is "a small program whose only purpose is to capture network traffic, while retaining advanced features like capturing to multiple files.
    - All three can write to a file. Wireshark's GUI can select which packets one wants to save. (tshark will record everything.)

tcpdump is a different, older, traffic capture application. It never had a GUI. And has a very different filter syntax, and capture packet format.

(Personal preference… I use tcpdump at the command line and for capture files. Then use Wireshark to look at the traffic in detail.)

The Capture Process

1. Select Capture | Interfaces
2. Select the interface on which packets need to be captured. (Usually be the interface where the Packet/s column is constantly changing, which would indicate the presence of live traffic).
3. Click the Start button to start the capture.
   a. The capture dialog should show the number of packets increasing. Avoid running any other applications while capturing, closing other browsers, etc.
4. Once the problem which is to be analyzed has been reproduced, click on Stop. May take a few seconds for Wireshark to display the packets captured.
5. Save the packet trace in the default format. File menu option > Save As.
   a. Default is pcap.ng


Wireshark Packet Capture Setup Process
START
Do you allowed to captured packet ?
Y
install Wireshark and packet capture support on the OS e.g On Windows install WInPcap
Open Wireshark GUI as non-root user, or user should have sufficient priviledge to run this software
Choose the live interface and put the interface in Promiscuous mode
is Wi-FI interface
Y
set the interface in Monitor Mode
N
Click on start Option to start capturing the pcaket
Use Wireshark features (filters/statistics/save) for further analysis
N
STOP

Quick Demo!

Hands-On Exercise

Task: Identify the bad actor within the SMTP traffic captured within the pcap.

1. Download the pcap file from the following link: [SMTP pcap file]
2. Open Wireshark and import the pcap file.
3. Apply filters to focus on SMTP traffic only.
4. Analyze the SMTP packets to identify any anomalies

Additional Resources

[Wireshark Documentation]

[Wireshark Wiki]

[HackTheBox Intro to Network Traffic Analysis]

[CTFs]

[Packet Analysis with Wireshark]

Pull up →


ACM X GDC PRESENTS
LAN PARTY
SUNDAY, APRIL 21
12:00 - 6:00 PM
LOCATED IN SSB 160
FREE PIZZA AND ICE CREAM!
MARIO KART TOURNAMENT
OPEN CONSOLES AND PC


ACM X UC MERCED GDC
MARIOKART TOURNAMENT
Sunday April 21st
1:00 PM
SSB 160
1st place $100 gift card of your choice
Link for registration

Sources

https://books.google.com/books?hl=en&lr=&id=UzzlCwAAQBAJ&oi=fnd&pg=PP1&dq=wireshark+network+analysis&ots=fkYKfOnCj3&sig=AvzzWOM9klY5a3ay1Yf4paZNwyc#v=onepage&q=wireshark%20network%20analysis&f=false

https://help.salesforce.com/s/articleView?id=000385344&type=1

https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=4509943

https://www.geeksforgeeks.org/wireshark-packet-capturing-and-analyzing/#

https://www.geeksforgeeks.org/introduction-to-wireshark/

https://www.wireshark.org/docs/

https://wiki.wireshark.org/SampleCaptures